

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI

EXECUTIVE SUMMARY

Lead AI Engineer & Data Architect (Google Developer Program Member), leveraging Google Antigravity to design and orchestrate neuro-symbolic autonomous multi-agent systems, Neo4j GraphRAG, and sovereign production MLOps. Specialized in deterministic architectures and verifiable AI trust. Founder of Archi Cam AI (official applicant for Google Africa Applied AI Lab), bridging civil engineering methodology, math rigor, and product vision.

FLAGSHIP AI PROJECTS

Archi Cam AI – *Agentic AI & 5D BIM SaaS*

- Official Applicant Google Africa Applied AI Lab. Built with Google Antigravity (Gemma 4 12B, Gemini, BAEI 91).
- Automated Excel BOQs in <45s (-99.2% time, MLflow $R^2=0.9872$) and 3D renders via Imagen 3 + ControlNet.

K1-MATHINFO (v3.0.0) – *Sovereign Multi-Agent AI & OKF Certification*

- Sovereign DMI system (Univ. of Ngaoundéré) engineered with Google Antigravity: 470 theses, 19 M1 projects, 1,366 nodes.
- 6-agent LangGraph network, hybrid retrieval (RRF $k=60$ + Cross-Encoder), SHA-256 No-LLM cert, 77/77 tests (100%).

Sovereign.BI Agentic – *Enterprise Security & Agentic BI*

- Sovereign natural language query engine for enterprise SQL/Graph databases (PostgreSQL pgvector, Neo4j, <5s latency).
- Enforces dynamic ABAC security guardrails and explainability certified via SHAP Sentinel game theory auditor.

Dataset Automator & VigieSahel – *MLOps Pipeline & Climate AI Impact*

- Dataset Automator: Built with Google Antigravity (Google Cloud Hackathon) with TabFM, BigQuery DataFrames, and EU AI Act.
- VigieSahel: Predictive ML for epidemic forecasting & crop sowing optimization (-35% loss, XGBoost $R^2>94\%$, Supabase).

PROFESSIONAL EXPERIENCE

AI Lead & Data Science Consultant (2025 – Present)

Independent Projects & Enterprises | Douala, CM

- Guiding enterprises in deploying sovereign private AI agents, Neo4j GraphRAG pipelines, and PostgreSQL databases.
- High-dimensional exploratory data analysis, predictive modeling, and executive interactive decision dashboards.

Aviation Security Officer (AVSEC) (2018 – Present)

CCAA (Cameroon Civil Aviation Authority)

- Operational threat assessment, secure access control, and regulatory compliance audits (ICAO Annex 17).
- Critical aviation security reporting and field emergency operational crisis response coordination.

EDUCATION & CERTIFICATIONS

Master of Science in Applied Artificial Intelligence (2025 – 2027) [In Progress]

University of Ngaoundéré

Excellence Curriculum (5 Blocs): Machine Learning & Bayesian Statistics, Data Engineering & Neo4j Knowledge Graphs, Computer Vision & Robotics/HCI, Cybersecurity & Blockchain, Production Sovereign MLOps Deployment.

Bachelor of Science in Civil Engineering (Building Option) (2015 – 2016)

ISTDI / IUC Douala

Structural calculations (BAEI 91), quantity surveying, and construction project management.

HONORS & APPLIED AI RECOGNITIONS

◆ **Google Africa Applied AI Lab (Accra, 2026):** Official candidacy supported by Archi Cam AI platform.

◆ **Google Cloud #AllThingsAgentic Hackathon:** Dataset Automator v4.0 built with Google Antigravity (TabFM, WIT, bigframes, MCT).



CONTACT & PROFILES

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📍 Douala / Ngaoundéré, CM

🌐 github.com/gervais-afk

🌐 linkedin.com/in/marie-gervais-koa

📧 devpost.com/magenel85

🏠 Google Developer Program

AI & LLM STACK

Google Antigravity IDE ■ ■ ■ ■ ■ ■ ■ ■

Google Gemma 4 (12B) ■ ■ ■ ■ ■ ■ ■ ■

Gemini 2.5 / 1.5 Pro ■ ■ ■ ■ ■ ■ ■ ■

Google TabFM (Tabular) ■ ■ ■ ■ ■ ■ ■ ■

LangGraph Multi-Agents ■ ■ ■ ■ ■ ■ ■ ■

Firebase Genkit ■ ■ ■ ■ ■ ■ ■ ■

Neo4j GraphRAG (N10S) ■ ■ ■ ■ ■ ■ ■ ■

DATA & GRAPHS

Neo4j / Cypher Graph ■ ■ ■ ■ ■ ■ ■ ■

PostgreSQL / pgvector ■ ■ ■ ■ ■ ■ ■ ■

Google BigQuery Data ■ ■ ■ ■ ■ ■ ■ ■

Supabase Realtime ■ ■ ■ ■ ■ ■ ■ ■

Pandas / NumPy ETL ■ ■ ■ ■ ■ ■ ■ ■

DEV & MLOPS

Python (3.11+ MLOps) ■ ■ ■ ■ ■ ■ ■ ■

FastAPI / Next.js 14 ■ ■ ■ ■ ■ ■ ■ ■

MLflow & Data Drift ■ ■ ■ ■ ■ ■ ■ ■

SHAP Sentinel Audit ■ ■ ■ ■ ■ ■ ■ ■

Docker & Streamlit ■ ■ ■ ■ ■ ■ ■ ■

IfcOpenShell (5D BIM) ■ ■ ■ ■ ■ ■ ■ ■

SECURITY & AUDIT

◆ Quorum 4-Eyes (KOA+AZIZ)

◆ OKF v0.2 SHA-256 No-LLM

◆ EU AI Act (RSASSA-PSS)

◆ AVSEC (ICAO Annex 17)

LANGUAGES

French (Native / Fluent)

English (Technical / Pro)

KEY ASSETS

◆ Dual Competence AI & Civil Eng.

◆ Aviation Security & Crisis (CCAA)

◆ Mathematical Rigor & Guardrails